

PGR Intelligence - Exceptional AI Product Vision

Detailed product roadmap for upgrading the existing PGR AI layer into an evidence-first, closed-loop intelligence system that is deeply aligned to the current PGR lifecycle and technical architecture.

Idea, value and product strategy	PGR Intelligence Programme
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Built around the existing PGR platform: lifecycle, supervision, progression, funding, thesis, documents, workflows, Pattern Lab and Ask PGR.

1. Executive summary

The current PGR platform already has a strong AI baseline: a rules-first Ask PGR assistant with confirmed actions, AI-ranked Weekly Review, grounded Meeting Briefs, Relationship Signals, Composer, and governed classical ML through Pattern Lab. The next stage should not be more isolated AI features. It should connect these assets into one intelligence layer that understands the research journey, explains the evidence, simulates options, prepares intervention and tracks what happened afterwards.

North-star product promise

Understand -> Predict -> Simulate -> Act - with evidence attached to every important conclusion and human confirmation on every consequential change.

Dimension	Current strength	Target exceptional state
Language interface	Ask PGR already queries and stages actions	Case Copilot maintains a working case, investigates across domains and prepares a coordinated action plan.
Prioritisation	Weekly Review ranks who needs attention	Intervention Queue explains the intervention type, suggested owner and reassessment trigger.
Prediction	Pattern Lab provides governed model probabilities	Risk Storyline + sensitivity show how the score changed and which inputs are worth investigating.
Evidence	Trace panel explains routing and source path	Evidence Ledger links every claim to rows, document passages, model versions and freshness.
Workflow	Tasks and confirmations already exist	Closed-loop intelligence records the intervention, completion and observed after-state.

2. Alignment to the existing PGR platform

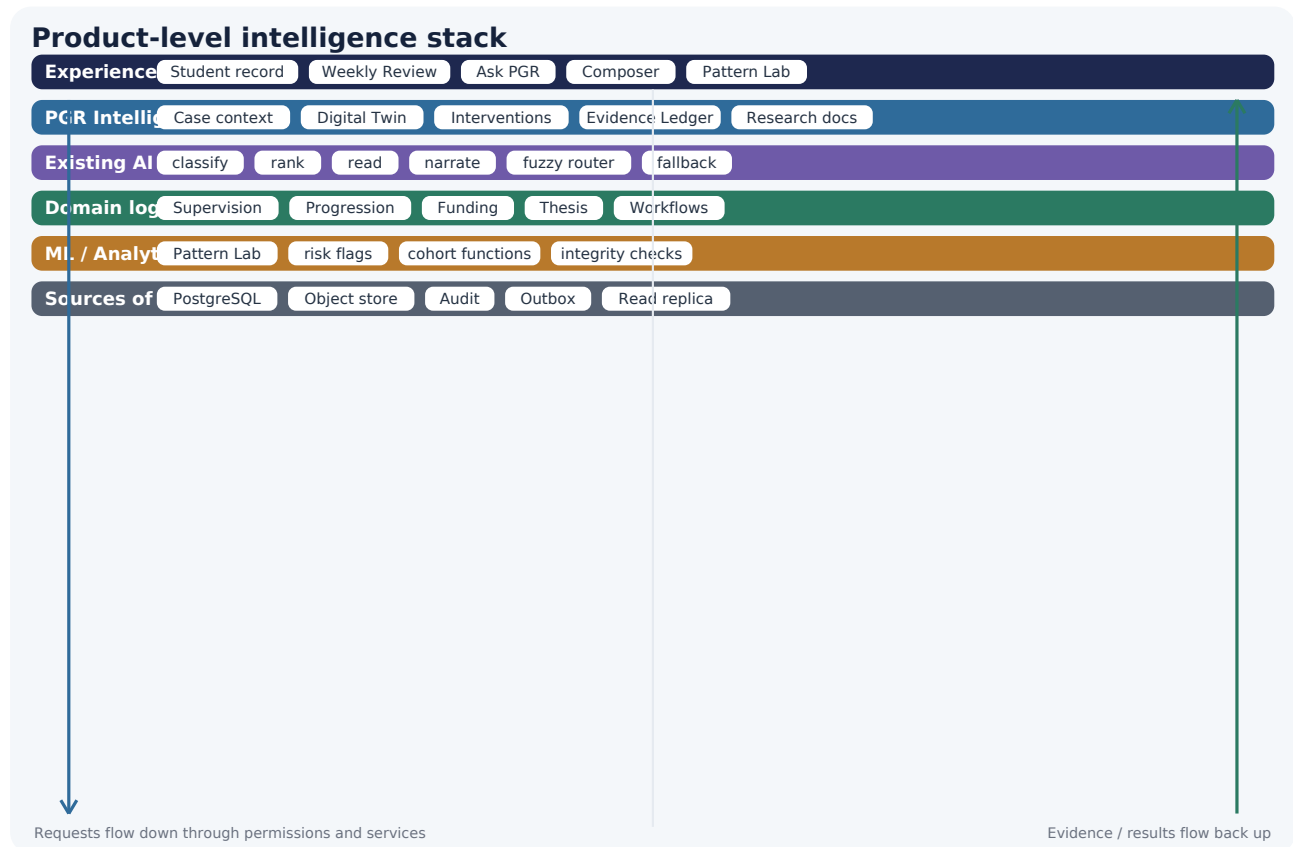
This roadmap deliberately builds on the current application rather than proposing a second platform. The student record already spans lifecycle changes, Pattern Lab prediction, supervisors, meetings, milestones, funding, thesis, documents and history. Workflows and tasks already create operational follow-through, and the student portal already exposes the same lifecycle actions with row scoping.

Existing platform capability	Why it matters for AI	Upgrade enabled
Single person / student journey	AI can reason over longitudinal context instead of disconnected module snapshots.	PGR Digital Twin, Risk Storyline, Case Copilot
Supervision + meetings	Provides cadence and agreed-action context.	Supervision Copilot, Engagement Trajectory
Progression milestones + panels + appeals	Provides formal deadlines and decision checkpoints.	Pressure-point detection, evidence mapping, bottleneck analysis
Funding lineage + integrity checks	Provides deterministic facts about coverage and gaps.	Funding-aware scenario previews and interventions
Thesis + corrections + documents	Provides research artefacts and formal examination state.	Research Change Radar and evidence workbench
Workflow engine + tasks	Provides a safe execution path for proposed actions.	Intervention Engine and staged multi-action plans
Pattern Lab governance	Provides governed predictive signals.	Trajectory + non-causal scenario sensitivity

Important architecture delta: the supplied deployment architecture restricts egress to SMTP and partner webhook targets, while the current AI inventory already uses Groq. Production network documentation should explicitly allow outbound HTTPS to the approved Groq endpoint or an institutional AI gateway. This is an alignment item, not a new product feature.

3. Product architecture - one intelligence layer, not many AI buttons

The target experience is a coherent intelligence layer shared by the student record, Weekly Review, Composer, Pattern Lab and Ask PGR. The same four actions should appear consistently: Explain, Evidence, Simulate and Prepare action.



Design invariant: rules calculate, ML predicts, LLM interprets/explains, humans decide. The exceptional experience comes from orchestration and evidence, not from giving the LLM authority over source-of-truth state.

4. PGR Digital Twin

A live, time-aware representation of the student journey that answers four questions: What is the current state? What is the next pressure point? What is blocking progress? What changes under a proposed scenario?

Dimension	Detailed design
Primary users	Supervisors, PGR administrators, research school leads
Current assets reused	Student record, lifecycle changes, milestones, funding lineage, tasks, Pattern Lab predictions
New capability	Derived StudentTwinSnapshot with state, trend, pressure points, dependencies and scenario diffs
Example	"If this student suspends for 12 weeks, which undecided milestones move, what happens to expected completion, and does the funding coverage still hold?"
Guardrail	Deterministic dates and funding calculations come from domain rules; ML scenario sensitivity is shown separately and labelled non-causal
Value	Reduces time spent assembling a case and makes cross-domain consequences visible before a request is raised
Moat	Requires PGR-specific lifecycle semantics and trusted longitudinal data, not just a general SIS chatbot
Success metrics	Case-understanding time, scenario completion rate, agreement with identified pressure point, number of prevented incomplete requests

5. AI Intervention Engine

Turns insight into a controlled, measurable operating loop: identify need -> recommend intervention -> stage actions -> confirm -> execute through existing workflows -> reassess later.

Dimension	Detailed design
Primary users	Supervisors and administrators working the Weekly Review Queue
Current assets reused	Weekly Review ranking, Meeting Brief, workflow/task engine, confirmation cards, audit trail
New capability	InterventionPlan containing intervention type, rationale, owner, action templates, dependencies, due-date source and reassessment date
Example	"Supervisor check-in" can stage a meeting task, generate the brief, create a review-evidence task and schedule reassessment - all visible before confirmation
Guardrail	The recommendation never bypasses current permissions or directly writes domain tables
Value	Shortens time from detection to appropriate action and distinguishes student issues from institutional blockers
Moat	Closed-loop intervention history compounds into institution-specific operational intelligence
Success metrics	Time-to-action, acceptance/edit rate, completion rate, reassessment outcome, duplicate-action prevention

6. Research Change Radar

Longitudinal semantic comparison of approved research artefacts. It highlights material changes in aims, methods, population, scope or assumptions and links each finding to evidence.

Dimension	Detailed design
Primary users	Supervisors, progression reviewers, administrators
Current assets reused	Documents on the student record, progression milestones, thesis/corrections state
New capability	Versioned document extraction, semantic diff, objective/method extraction and ChangeFinding objects with exact source locators
Example	Proposal: qualitative interviews with 40 participants. Annual review: mixed methods with 120 participants. Radar flags methodology and population changes for human review.
Guardrail	A detected change is not automatically a compliance or ethics breach; the system says "review implication" until the relevant rule/human decision applies
Value	Reviewers focus on what changed instead of rereading entire documents
Moat	Deeply PGR-specific research lifecycle intelligence
Success metrics	Accepted change-findings, false-positive rate, reviewer time, evidence locator accuracy

7. Institutional Memory

Permission-aware retrieval of comparable resolved cases so staff can see how similar situations were handled and what evidence was commonly requested.

Dimension	Detailed design
Primary users	Senior PGR administrators, Registry, school operations
Current assets reused	Audit/history, workflow outcomes, lifecycle requests, funding/progression context
New capability	Governed case feature builder + hybrid filters/similarity + de-identified precedent summaries
Example	"Show previous part-time extension cases where funding ended before the revised completion date."
Guardrail	Frequency of prior outcomes must never become an automatic recommendation; comparable cases are context, not authority
Value	Faster unusual-case preparation, more consistent evidence requests, better staff onboarding
Moat	Institution-specific operational precedent becomes more valuable with time
Success metrics	Case preparation time, comparable-case opens, consistency of evidence requested, user challenge rate

8. Policy Compiler

A controlled policy-to-configuration workspace: compare approved policy versions, propose allow-listed system changes, identify affected cohorts and simulate impact before publishing.

Dimension	Detailed design
Primary users	Registry / policy owners, system administrators, governance
Current assets reused	Institution settings, progression rules, workflows, affected-record queries
New capability	Clause extraction, change detection, constrained RuleCandidate / ConfigCandidate schema, policy impact simulation
Example	Annual review window changes from 12 to 10 months. The compiler proposes the configuration delta and shows how many undecided milestones would move.
Guardrail	Never compile free policy prose directly into executable code or expressions; only allow-listed settings/rules can be proposed
Value	Reduces manual interpretation risk and makes policy change impact testable before go-live
Moat	Connects policy text to live PGR configuration and cohort impact, beyond policy search
Success metrics	Time from policy approval to configuration, manual corrections, impact simulation accuracy, rollback incidents

9. Ask PGR -> Case Copilot

Ask PGR should evolve from a powerful natural-language command layer into a case-oriented copilot without losing the current deterministic router, row scoping, confirmation cards or traceability. The upgrade is not "more chat"; it is persistent working context and multi-step orchestration.

Current behaviour	Upgrade	User value
Resolve intent / entities / time window	Maintain a typed CaseContext across turns	Follow-up questions preserve the student, cohort, filters, issue and pending action plan.
Answer grounded questions	Investigate across multiple authorised domains	"Why is this student in Weekly Review?" can combine milestone, meeting, funding and prediction evidence.
Stage one confirmed action	Prepare a multi-action plan	One confirmation card can contain several proposed commands, each editable.
Trace matched vocabulary / LLM path	Attach Evidence Ledger entries	Users can move from answer -> evidence -> source record.
Fallback to deterministic path	Fallback preserves case shell and manual action path	Groq outage degrades quality, not core workflow availability.

Exceptional detail: Case Copilot should be able to stop and say what it cannot determine. Example: "Funding is approved in PGR, but Finance inbound confirmation is not available in the documented architecture, so I cannot confirm payment receipt."

10. Weekly Review -> Intervention Queue

The upgraded queue should separate the reason for attention from the recommended response. This reduces the risk of treating every alert as a student problem when the blocking step may belong to a supervisor, panel, Registry or Finance.

Layer	What the queue shows	Example
Priority	Deterministic urgency + governed model signals	Review due in 18 days; 82 days since meeting
Cause category	Academic / supervisory / administrative / institutional	Institutional - panel confirmation waiting
Suggested intervention	A reviewable action template, not an autonomous action	Supervisor check-in + review evidence preparation
Owner	Resolved from existing roles/workflow state	Supervisor: Dr Elena Ford
Reassessment	Date/event when the case should be re-evaluated	14 days or after meeting completion
Outcome	Observed before/after state; no causal claim	Attention HIGH -> MEDIUM after actions completed

11. Meeting Brief -> Supervision Copilot

Turn the Meeting Brief into a before/during/after loop. The exceptional value is continuity of agreed actions and dependencies, not AI-written meeting prose.

Stage	AI behaviour	Safety / quality rule
Before	Summarise changes since last meeting, previous commitments, blockers, upcoming milestones and funding/thesis context.	Only authorised evidence; cite source rows/documents.
During	Structure shorthand notes into proposed decisions, actions, owners and dependencies.	Never invent deadlines or owners not supported by notes/context.
After	Generate a review screen before saving the meeting record and tasks.	Human confirms each extracted commitment.
Next meeting	Carry confirmed open actions and dependencies into the next brief.	Do not treat blocked student work as student non-compliance.

12. Relationship Signal -> Engagement Trajectory

Keep the lightweight status badge, but make the deeper view a time series of observed supervision engagement. The system should describe recorded interaction patterns rather than make claims about the quality of a human relationship.

Upgrade	Detailed behaviour
Trajectory	Show monthly/term movement in contact status, meeting recency and completion of agreed supervision actions.
Reason codes	Each change cites the evidence that moved the signal: meeting logged, expected interval missed, action completed, explicit context override.
Context override	Allow authorised users to record fieldwork, approved leave or missing historical meetings so the signal can be interpreted correctly.
Fallback	Rule-based contact health remains available when the LLM is off; LLM only improves narrative/classification at the edges.
Language	Prefer "Supervision engagement" / "No recent contact recorded" over "relationship is poor".

13. Composer -> Institutional Investigator

Composer should keep its fixed catalogue of trusted data functions, but the functions should return drill paths and contributing record IDs so the AI can perform a controlled investigation rather than only assemble blocks.

Investigation step	Example	Engine
Question	Why are Biology annual reviews taking longer this year?	LLM interprets question and chooses catalogue functions
Measure	Median duration increased 22 -> 31 days	Deterministic function result
Locate change	Panel -> decision stage +7.3 days	Deterministic stage-duration function
Find contributors	19 cases explain most of increase	Deterministic record contribution query
Explain	Increase is concentrated after panel convening	LLM narrative over returned facts
Operationalise	Save monthly monitor / open contributing cases	Existing dashboard / workflow routes

14. Pattern Lab -> Intervention Lab + Risk Storyline

Pattern Lab is already the most technically mature AI/ML surface. The product upgrade should make it easier to understand change over time and safer to explore model sensitivity without implying causation.

Upgrade	Detailed behaviour	Boundary
Score history	Persist each mature prediction with model version and timestamp.	Do not compare scores across materially different model versions without marking the break.
Risk Storyline	Explain what changed since the prior score using stored driver snapshots.	Driver change explains model movement, not real-world causality.
Sensitivity explorer	Recalculate score under controlled input changes.	Permanent label: model sensitivity, not predicted intervention effect.
Model health in context	Show PSI, matured AUC / calibration status near the prediction.	If model health is poor, suppress strong recommendations and route to governance review.
Intervention link	Route investigable factors to Evidence / Prepare intervention.	No automatic intervention from model probability alone.

15. AI Evidence Ledger + "know what you do not know"

The Evidence Ledger is the trust layer that should sit beneath every other capability. It should show verified facts, predictions, document findings, missing sources and freshness in one uniform structure.

Evidence class	Examples	How displayed
Deterministic fact	Milestone due date, funding end, last meeting, task status	Verified with source ID and current value
Model signal	Milestone drift 67%, model version PL-MD-v7	Prediction with model health and timestamp
Document finding	Methodology changed between Proposal v2 and Review v1	Needs review with exact passage links
Missing evidence	Finance payment confirmation unavailable	Explicitly limits conclusion
User context	Fieldwork until October	Authorised context override with actor + timestamp

Avoid fake "92% confidence" badges. A better UI is: Evidence coverage = complete / partial; Missing information = none / listed; Model health = healthy / review; Human review = required / completed.

16. Proactive intelligence - move from "ask me" to "I noticed"

Use the existing worker to run scheduled checks for material changes. The key is restraint: proactive AI should surface only meaningful deltas, systemic patterns or governance issues, with suppression rules to avoid alert fatigue.

Alert type	Example	Route
Systemic bottleneck	14 annual reviews within 30 days have no panel confirmed; up from 3 last week.	Open Composer investigation / affected cases
Funding coverage	8 students now have completion after recorded funding end; 5 changed after lifecycle updates.	Open Funding integrity / intervention queue
Model health	Production milestone-drift model matured AUC / PSI indicates review.	Open Pattern Lab governance
Research change	A newly uploaded review document contains a material methodology change.	Open Change Radar comparison
Policy impact	A proposed configuration change would make 14 open workflows overdue.	Open Policy Compiler simulation

17. Value by persona and measurable outcomes

The roadmap should be justified through operating outcomes, not "AI engagement" metrics alone.

Persona	Most valuable upgrades	Primary measurable outcome
Supervisor	Supervision Copilot, Digital Twin, Intervention Queue	Preparation time; action follow-through; fewer missed review dependencies
PGR administrator	Case Copilot, Institutional Memory, Policy Compiler	Case handling time; consistency; fewer incomplete requests
Research student	Evidence-backed My Next Move; private thesis/viva support later	Clarity of next steps; fewer avoidable admin queries
Executive / school lead	Institutional Investigator, proactive systemic alerts	Time to identify bottlenecks; unresolved queue age
Registry / governance	Evidence Ledger, Policy Compiler, Pattern Lab health	Traceability; policy implementation quality; governed model use

18. Recommended delivery sequence

The build order should maximise reuse of foundations and minimise architecture churn.

Release	Foundations	User-visible upgrades	Why now
R1 - Trust + context	EvidenceClaim, CaseContext, score history, action-plan schema	Case Copilot, Risk Storyline, Evidence Drawer	Build shared contracts before new features multiply
R2 - Closed loop	InterventionPlan, outcome snapshots, engagement events	Intervention Queue, Supervision Copilot, Engagement Trajectory	Extends features you already have with modest data-model changes
R3 - Simulation	Side-effect-free scenario engine + sensitivity runner	PGR Digital Twin scenario preview	High demo value once trust layer is solid
R4 - Documents	Versioned extraction, locators, semantic comparison	Research Change Radar + evidence mapping	Requires stronger document pipeline
R5 - Institutional moat	Case similarity, policy mapping schema	Institutional Memory + Policy Compiler	Highest compounding value, but needs mature data governance

Recommended first implementation: build EvidenceClaim + CaseContext + score-history storage, then upgrade Ask PGR, Weekly Review and Meeting Brief before starting Research Change Radar. This gives immediate product value and creates the foundations used by every later capability.

19. Source basis, alignment notes and open assumptions

This document is grounded in the supplied PGR User Manual v2, the supplied System Architecture & Deployment Sizing document, and the current AI inventory provided in the conversation. Proposed capabilities are explicitly described as upgrades; they are not assumed to exist already.

- Current platform anchors used: continuous person/student journey; student marquee record; supervision; progression; funding/funding integrity; thesis; workflows/tasks; student portal; Pattern Lab; Ask PGR; audit/outbox.
- Current technical anchors used: Next.js 14 frontend; FastAPI API; APScheduler worker; PostgreSQL primary; optional read replica; object store; existing rate limit on /assistant; private API/DB network boundaries.
- Current AI anchors used from your inventory: Groq gpt-oss-120b/20b; typed app/ai classify/rank/read/narrate contracts; row-scoped prompts; model-off integration test; deterministic fuzzy router and cohort query engine.
- Not supplied: PGR_Platform_Critical_Gaps.pdf. Any gap described there may change implementation priority and should be reviewed before production commitment.
- New data structures suggested here - CaseContext, EvidenceClaim, TwinSnapshot, InterventionPlan, ChangeFinding, PolicyProposal - are proposed derived/intelligence objects, not replacements for existing source-of-truth domain tables.